Beomyoung Kim 김범영



Research Scientist — NAVER Cloud (Image Vision)

Self-Improving Agents · Multimodal AI · Computer Vision

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SUMMARY

I am a Research Scientist at NAVER Cloud and a Ph.D. student at KAIST. My current research focuses on **self-improving agents** — meta-agents that use execution experience to diagnose limitations and iteratively improve agent harnesses with frozen model weights. I currently focus on **visual agents**, aiming to develop broadly applicable methods for agent improvement.

My earlier work spans label-efficient and continual **visual recognition** — segmentation, detection, and matting — and **grounded VLMs and visual reasoning agents**. This background in perception and diagnostic reasoning informs my current research. I have published **14+ papers** at CVPR, ICCV, ECCV, NeurIPS, and AAAI (608 citations, h-index 10), including an **ICCV 2025 Highlight**, and brought research into production through a zero-shot image-matting foundation model for CLOVA-X image editing, a foreground-segmentation API, and a face-authentication system serving millions of users.

HIGHLIGHTS

- **7 first-author papers** at top-tier venues — CVPR · ICCV · ECCV · NeurIPS
- **5+ years** as a Research Scientist at NAVER Cloud with a proven **research → product** track record
- **Ph.D. candidate** pursued in parallel with full-time research
- **Current research** — Self-Improving Agents & Automated Agent Design

EXPERIENCE

Research Scientist — NAVER Cloud, Image Vision

Jan 2021 – Present

Seongnam, Korea · formerly NAVER CLOVA, reorganized into NAVER Cloud

- **Self-Improving Agents** (ongoing) — Developing meta-agents that analyze execution trajectories, diagnose limitations, and iteratively propose, apply, and evaluate harness updates across prompts and instructions, workflows and tool use, code and agent architecture, and memory and skills, with all model weights frozen. Currently focused on visual agents, with the goal of generalizing the improvement process across agent systems.
- **Grounded VLMs & Visual Reasoning Agents** — Researched VLMs grounded in pixel- and region-level evidence and training-free agentic program synthesis that composes specialist vision models into executable workflows for spatial and temporal reasoning. Developed failure diagnosis and targeted program repair for 3D spatial reasoning.
- Created **ZIM**, a promptable zero-shot image-matting foundation model (**ICCV 2025 Highlight**), by improving SAM through architecture, loss, and data-pipeline design on an ~1M-image curated dataset; open-sourced with a public demo and integrated into the CLOVA-X Image Editing service.

- Developed the model and dataset for a **foreground-segmentation API** now served internally, outperforming Photoroom, Remove.bg, and Adobe segmentation APIs in internal evaluation.
- Built the face anti-spoofing model behind **FaceSign**, a face-authentication system replacing payment and employee-badge tagging, for this government-certified service.
- Independently developed an efficient **on-device (mobile-CPU) human-segmentation model**, deployed via ONNX-based in-house serving, achieving ~10ms inference with robust quality under tight mobile compute budgets.
- Advanced **label-efficient and continual segmentation**, producing first-author CVPR papers — ECLIPSE (continual panoptic), PointWSSIS (point-supervised instance), and BESTIE (weakly-supervised instance) — while developing annotation-efficient methods and widely adopted open-source codebases.

Research Internships

2018 – 2020

- **NAVER CLOVA Visual AI (FACE)**, 2020 — face recognition / understanding research.
- **Hyundai Mobis, Autonomous Driving Advanced Development**, 2019 — perception for autonomous driving.
- **NAVER CLOVA Vision (OCR)**, 2018 — optical character recognition research.

EDUCATION

Ph.D., Kim Jaechul Graduate School of AI — KAIST

2022 – Present

MLAI Lab, advised by Prof. Sung Ju Hwang (in parallel with full-time work)

M.S., Electrical Engineering (Future Vehicle) — KAIST

2019 – 2021

SIIT Lab, advised by Prof. Junmo Kim

B.S., Information and Communication Engineering — Inha University

2013 – 2019

UNDER REVIEW

UNDER REVIEW · ICLR 2027 **Visual Reasoning Agents.** A diagnostic framework for 3D spatial reasoning that turns silent perception failures in visual program synthesis into typed diagnoses, driving targeted program repair to rival frontier VLMs without task-specific training.

SELECTED PUBLICATIONS

(★ = first author · full list on Scholar)

1. **Less Supervision, Better Generalization: Weakly Supervised Fake Region Localization in Diffusion-Edited Images.**
   Junhee Lee, JeonDongHyeon, Taeoh Kim, **Beomyoung Kim**, MyeongAh Cho.
   NeurIPS 2026 · paper · code coming soon

2. ★ **Learning from Adversity: Semantic-Aware Mask Refinement through Adversarial Perturbation.**
   Beomyoung Kim, Sung Ju Hwang.
   ECCV 2026 · paper · code · project

3. ★ **ZIM: Zero-Shot Image Matting for Anything.**
   Beomyoung Kim, Chanyong Shin, Joonhyun Jeong, Hyungsik Jung, Se-Yun Lee, Sewhan Chun, Dong-Hyun Hwang, Joonsang Yu.
   ICCV 2025 Highlight · paper · code · project

4. ★ **Towards Label-Efficient Human Matting: A Simple Baseline for Weakly Semi-Supervised Trimap-Free Human Matting.**
   Beomyoung Kim, Myeong Yeon Yi, Joonsang Yu, Young Joon Yoo, Sung Ju Hwang.
   arXiv 2024 · paper · code

5. ★ **Rethinking Saliency-Guided Weakly-Supervised Semantic Segmentation.**
   Beomyoung Kim, Donghyeon Kim, Sung Ju Hwang.
   arXiv 2024 · paper · code

6. ★ **ECLIPSE: Efficient Continual Learning in Panoptic Segmentation with Visual Prompt Tuning.**
   Beomyoung Kim, Joonsang Yu, Sung Ju Hwang.
   CVPR 2024 · paper · code

7. **EResFD: Rediscovery of the Effectiveness of Standard Convolution for Lightweight Face Detection.**
   Joonhyun Jeong, **Beomyoung Kim**, Joonsang Yu, Youngjoon Yoo.
   WACV 2024 · paper · code

8. ★ **The Devil is in the Points: Weakly Semi-Supervised Instance Segmentation via Point-Guided Mask Representation.**
   Beomyoung Kim, Joonhyun Jeong, Dongyoon Han, Sung Ju Hwang.
   CVPR 2023 · paper · code

9. ★ **Beyond Semantic to Instance Segmentation: Weakly-Supervised Instance Segmentation via Semantic Knowledge Transfer and Self-Refinement.**
   Beomyoung Kim, Youngjoon Yoo, Chaeeun Rhee, Junmo Kim.
   CVPR 2022 · paper · code

10. **Learning Features with Parameter-Free Layers.**
    Dongyoon Han, Youngjoon Yoo, **Beomyoung Kim**, Byeongho Heo.
    ICLR 2022 · paper · code

11. ★ **TricubeNet: 2D Kernel-Based Object Representation for Weakly-Occluded Oriented Object Detection.**
    Beomyoung Kim, Janghyeon Lee, Sihaeng Lee, Doyeon Kim, Junmo Kim.
    WACV 2022 · paper · code

12. ★ **SSUL: Semantic Segmentation with Unknown Label for Exemplar-based Class-Incremental Learning.**
    Sungmin Cha*, **Beomyoung Kim***, Youngjoon Yoo, Taesup Moon *(* equal contribution)*.
    NeurIPS 2021 · paper · code

13. ★ **Discriminative Region Suppression for Weakly-Supervised Semantic Segmentation.**
    Beomyoung Kim, Sangeun Han, Junmo Kim.
    AAAI 2021 · paper · code

14. ★ **3D Point Cloud Upsampling and Colorization using GAN.**
    Beomyoung Kim, Sangeun Han, Eojindl Yi, Junmo Kim.
    MIWAI 2021 · paper

15. **Fully Automated Valet Parking System Based on Infrastructure Sensing.**
    Hyunjee Ryu, **Beomyoung Kim**, Heecheol Yoo, Jungwon Lee.
    RiTA 2020 · paper

HONORS & AWARDS

2025 ICCV 2025 **Highlight** (top ~3% of accepted papers) — ZIM

ACADEMIC SERVICE — REVIEWER

2026 CVPR · ECCV · NeurIPS · ICLR · TPAMI

2025 CVPR · ICCV · NeurIPS · ICLR · TMLR

2024 CVPR · ECCV · NeurIPS · ICLR · AAAI

2023 CVPR · ICCV · NeurIPS · WACV

INVITED TALKS

2025 Centum Digital Week — "Next Code 2025: Beyond AI, Into Agents"

2022 Jinhaksa Catch Career-Con — AI Research Engineer career

2021 NeurIPS 2021 Social: ML in Korea — SSUL

2024 TEAM NAVER DAN 24 — CLOVA-X Image Editing

2022 Inha University — Weakly-Supervised Instance Segmentation

TECHNICAL SKILLS

Research areas: Self-Improving Agents · Automated Agent Design · Visual Reasoning Agents · Multimodal / Vision-Language Models · Vision Foundation Models · Image Segmentation, Detection & Matting · Label-Efficient & Continual Learning

Frameworks & tools: PyTorch · TensorFlow · large-scale / distributed training

Programming: Python · C++ · C · Java

LANGUAGES

Korean (native) · English (intermediate — conversational working proficiency)